

Quantium Retail Analytics Internship

Task 2: Experimentation and Uplift Testing

Objective: Evaluate whether the new chip layout implemented in trial stores (77, 86, 88) resulted in a significant increase in sales compared to similar control stores.

Key Metrics:

- Total Sales
- Number of Customers
- Transactions per Customer

In [290...

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind

sns.set_style("whitegrid")
```

In [291...

```
data = pd.read_csv(r"D:\Forage\QVI_data.csv")
data.head()
```

Out[291...

	LYLTY_CARD_NBR	DATE	STORE_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_QT
0	1000	2018-10-17	1	1	5	Natural Chip Compny SeaSalt175g	
1	1002	2018-09-16	1	2	58	Red Rock Deli Chikn&Garlic Aioli 150g	
2	1003	2019-03-07	1	3	52	Grain Waves Sour Cream&Chives 210G	
3	1003	2019-03-08	1	4	106	Natural ChipCo Hony Soy Chckn175g	
4	1004	2018-11-02	1	5	96	WW Original Stacked Chips 160g	

◀

▶

In [292...

```
data.shape
```

Out[292...] (264834, 12)

In [293...] data.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264834 entries, 0 to 264833
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   LYLTY_CARD_NBR         264834 non-null int64
1   DATE                   264834 non-null object
2   STORE_NBR              264834 non-null int64
3   TXN_ID                 264834 non-null int64
4   PROD_NBR               264834 non-null int64
5   PROD_NAME              264834 non-null object
6   PROD_QTY               264834 non-null int64
7   TOT_SALES              264834 non-null float64
8   PACK_SIZE              264834 non-null int64
9   BRAND                  264834 non-null object
10  LIFESTAGE               264834 non-null object
11  PREMIUM_CUSTOMER       264834 non-null object
dtypes: float64(1), int64(6), object(5)
memory usage: 24.2+ MB
```

In [294...] data.describe()

Out[294...]

	LYLTY_CARD_NBR	STORE_NBR	TXN_ID	PROD_NBR	PROD_QTY
count	2.648340e+05	264834.000000	2.648340e+05	264834.000000	264834.000000
mean	1.355488e+05	135.079423	1.351576e+05	56.583554	1.905813
std	8.057990e+04	76.784063	7.813292e+04	32.826444	0.343436
min	1.000000e+03	1.000000	1.000000e+00	1.000000	1.000000
25%	7.002100e+04	70.000000	6.760050e+04	28.000000	2.000000
50%	1.303570e+05	130.000000	1.351365e+05	56.000000	2.000000
75%	2.030940e+05	203.000000	2.026998e+05	85.000000	2.000000
max	2.373711e+06	272.000000	2.415841e+06	114.000000	5.000000



In [295...] data.isnull().sum()

```
Out[295...] LYLTY_CARD_NBR      0
            DATE        0
            STORE_NBR   0
            TXN_ID      0
            PROD_NBR    0
            PROD_NAME    0
            PROD_QTY     0
            TOT_SALES    0
            PACK_SIZE    0
            BRAND        0
            LIFESTAGE    0
            PREMIUM_CUSTOMER 0
            dtype: int64
```

```
In [296...] # Convert the 'DATE' column to datetime format
data["DATE"] = pd.to_datetime(data["DATE"])
data["DATE"].head()
```

```
Out[296...] 0    2018-10-17
            1    2018-09-16
            2    2019-03-07
            3    2019-03-08
            4    2018-11-02
            Name: DATE, dtype: datetime64[ns]
```

```
In [297...] # Extract the month from the 'DATE' column
data["MONTH"] = data["DATE"].dt.to_period("M")
data["MONTH"].head()
```

```
Out[297...] 0    2018-10
            1    2018-09
            2    2019-03
            3    2019-03
            4    2018-11
            Name: MONTH, dtype: period[M]
```

```
In [298...] # Extract the month as a separate column month_id
data["MONTH_ID"] = data["DATE"].dt.month
data["MONTH_ID"].head()
```

```
Out[298...] 0    10
            1     9
            2     3
            3     3
            4    11
            Name: MONTH_ID, dtype: int64
```

```
In [299...] data["STORE_NBR"].unique()
data[data["STORE_NBR"].isin([77,86,88])].head()
```

Out[299...

	LYLTY_CARD_NBR	DATE	STORE_NBR	TXN_ID	PROD_NBR	PROD_NAME
73365	77000	2019-03-28	77	74911	18	Cheetos Chs & Bacon Balls 190g
73366	77000	2019-04-13	77	74912	69	Smiths Chip Thinly S/Cream&Onion 175g
73367	77000	2018-09-26	77	74910	36	Kettle Chilli 175g
73368	77001	2019-02-27	77	74913	7	Smiths Crinkle Original 330g
73369	77001	2019-01-21	77	74914	9	Kettle Tortilla ChpsBtroot&Ricotta 150g

In [300...

```
# Group the data by store and month, and calculate total sales, number of customers, and transactions
store_month = data.groupby(["STORE_NBR", "MONTH_ID"]).agg(
    tot_sales=("TOT_SALES", "sum"),
    customers=("LYLTY_CARD_NBR", "nunique"),
    transactions=("TXN_ID", "nunique")
).reset_index()

store_month["txn_per_customer"] = store_month["transactions"] / store_month["customers"]

store_month.head(13)
```

Out[300...

	STORE_NBR	MONTH_ID	tot_sales	customers	transactions	txn_per_customer
0	1	1	154.8	35	36	1.028571
1	1	2	225.4	52	55	1.057692
2	1	3	192.9	45	49	1.088889
3	1	4	192.9	42	43	1.023810
4	1	5	221.4	46	51	1.108696
5	1	6	174.1	42	42	1.000000
6	1	7	206.9	49	52	1.061224
7	1	8	176.1	42	43	1.023810
8	1	9	278.8	59	62	1.050847
9	1	10	188.1	44	45	1.022727
10	1	11	192.6	46	47	1.021739
11	1	12	189.6	42	47	1.119048
12	2	1	162.8	43	45	1.046512

In [301...

```
# Analyze the number of months each store has data for
```

```
store_month.groupby("STORE_NBR")["MONTH_ID"].nunique().value_counts()
```

```
Out[301...] 12    260  
            2      5  
            11     3  
            1      3  
            3      1  
            Name: MONTH_ID, dtype: int64
```

```
In [302...] store_month.shape
```

```
Out[302...] (3169, 6)
```

```
In [303...] store_month[store_month["STORE_NBR"].isin([77,86,88])]
```

Out[303...

	STORE_NBR	MONTH_ID	tot_sales	customers	transactions	txn_per_customer
880	77	1	204.40	35	39	1.114286
881	77	2	235.00	45	45	1.000000
882	77	3	278.50	50	55	1.100000
883	77	4	263.50	47	48	1.021277
884	77	5	299.30	55	56	1.018182
885	77	6	264.70	41	42	1.024390
886	77	7	296.80	51	55	1.078431
887	77	8	255.50	47	48	1.021277
888	77	9	225.20	42	44	1.047619
889	77	10	204.50	37	38	1.027027
890	77	11	245.30	41	44	1.073171
891	77	12	267.30	46	48	1.043478
977	86	1	841.40	94	129	1.372340
978	86	2	913.20	107	138	1.289720
979	86	3	1026.80	115	140	1.217391
980	86	4	848.20	105	126	1.200000
981	86	5	889.30	104	128	1.230769
982	86	6	838.00	98	118	1.204082
983	86	7	892.20	99	126	1.272727
984	86	8	764.05	94	110	1.170213
985	86	9	914.60	103	128	1.242718
986	86	10	948.40	109	138	1.266055
987	86	11	918.00	100	125	1.250000
988	86	12	841.20	98	120	1.224490
1001	88	1	1266.40	117	144	1.230769
1002	88	2	1370.20	124	153	1.233871
1003	88	3	1477.20	134	169	1.261194
1004	88	4	1439.40	128	162	1.265625
1005	88	5	1308.25	128	154	1.203125
1006	88	6	1354.60	121	148	1.223140
1007	88	7	1310.00	129	153	1.186047
1008	88	8	1323.80	131	158	1.206107
1009	88	9	1423.00	124	157	1.266129

	STORE_NBR	MONTH_ID	tot_sales	customers	transactions	txn_per_customer
1010	88	10	1352.40	123	155	1.260163
1011	88	11	1382.80	130	156	1.200000
1012	88	12	1325.20	126	148	1.174603

```
In [304... pretrial = store_month[store_month["MONTH_ID"].isin([7,8,9,10,11,12,1])]
pretrial.head(10)
```

```
Out[304...
```

	STORE_NBR	MONTH_ID	tot_sales	customers	transactions	txn_per_customer
0	1	1	154.8	35	36	1.028571
6	1	7	206.9	49	52	1.061224
7	1	8	176.1	42	43	1.023810
8	1	9	278.8	59	62	1.050847
9	1	10	188.1	44	45	1.022727
10	1	11	192.6	46	47	1.021739
11	1	12	189.6	42	47	1.119048
12	2	1	162.8	43	45	1.046512
18	2	7	150.8	39	41	1.051282
19	2	8	193.8	39	43	1.102564

```
In [305... pretrial.groupby("STORE_NBR")["MONTH_ID"].nunique().value_counts()
```

```
Out[305... 7    260
1         6
6         3
2         2
Name: MONTH_ID, dtype: int64
```

```
In [306... # exact trial store data for store 77
trial77 = pretrial[pretrial["STORE_NBR"] == 77]
trial77
```

```
Out[306...
```

	STORE_NBR	MONTH_ID	tot_sales	customers	transactions	txn_per_customer
880	77	1	204.4	35	39	1.114286
886	77	7	296.8	51	55	1.078431
887	77	8	255.5	47	48	1.021277
888	77	9	225.2	42	44	1.047619
889	77	10	204.5	37	38	1.027027
890	77	11	245.3	41	44	1.073171
891	77	12	267.3	46	48	1.043478

```
In [307... # Calculate the correlation between store 77 and all other stores
corr_dict = {}

for store in pretrial["STORE_NBR"].unique():

    if store != 77:

        candidate = pretrial[pretrial["STORE_NBR"] == store]

        merged = pd.merge(
            trial77,
            candidate,
            on="MONTH_ID",
            suffixes=("_trial", "_control")
        )

        corr = merged["tot_sales_trial"].corr(merged["tot_sales_control"])

        corr_dict[store] = corr

corr_df = pd.DataFrame.from_dict(corr_dict, orient="index", columns=["correlation"])
corr_df.sort_values("correlation", ascending=False).head(10)
```

```

c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)

```

Out[307...

	correlation
31	1.000000
11	1.000000
71	0.914106
233	0.903774
119	0.867664
17	0.842668
3	0.806644
41	0.783232
50	0.763866
157	0.735893

In [308...

```

# Select a control store for comparison
control77 = 233

```

```

plot_df = store_month[store_month["STORE_NBR"].isin([77, control77])]

for store in [77, control77]:

    subset = plot_df[plot_df["STORE_NBR"] == store]

    plt.plot(subset["MONTH_ID"], subset["tot_sales"], label=f"Store {store}")

plt.title("Trial Store vs Control Store Sales")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.legend()
plt.show()

```



Control Store Selection – Store 77

To evaluate the trial store performance, we first identified a suitable control store using correlation analysis on pre-trial monthly sales.

Store 233 showed a high correlation (0.90) with Trial Store 77 during the pre-trial period, indicating that both stores followed a similar sales pattern before the trial began.

A visual comparison of monthly sales trends further confirms that Store 233 closely mirrors the sales behaviour of Store 77. Therefore, Store 233 was selected as the control store for Trial Store 77.

In [309...

```

trial_period = store_month[store_month["MONTH_ID"].isin([2,3,4])]
trial_period.head()

```

Out[309...]

	STORE_NBR	MONTH_ID	tot_sales	customers	transactions	txn_per_customer
1	1	2	225.4	52	55	1.057692
2	1	3	192.9	45	49	1.088889
3	1	4	192.9	42	43	1.023810
13	2	2	139.4	29	32	1.103448
14	2	3	192.1	43	46	1.069767

In [310...]

```
trial_store = trial_period[trial_period["STORE_NBR"] == 77]
trial_store
```

Out[310...]

	STORE_NBR	MONTH_ID	tot_sales	customers	transactions	txn_per_customer
881	77	2	235.0	45	45	1.000000
882	77	3	278.5	50	55	1.100000
883	77	4	263.5	47	48	1.021277

In [311...]

```
control_store = trial_period[trial_period["STORE_NBR"] == 233]
control_store
```

Out[311...]

	STORE_NBR	MONTH_ID	tot_sales	customers	transactions	txn_per_customer
2700	233	2	244.0	45	47	1.044444
2701	233	3	199.1	40	41	1.025000
2702	233	4	158.6	30	32	1.066667

In [312...]

```
trial_sales = trial_store["tot_sales"].sum()
control_sales = control_store["tot_sales"].sum()

print("Trial store sales:", trial_sales)
print("Control store sales:", control_sales)
```

Trial store sales: 777.0
Control store sales: 601.7

In [313...]

```
# Compare trial store with control store
compare = store_month[store_month["STORE_NBR"].isin([77,233])]

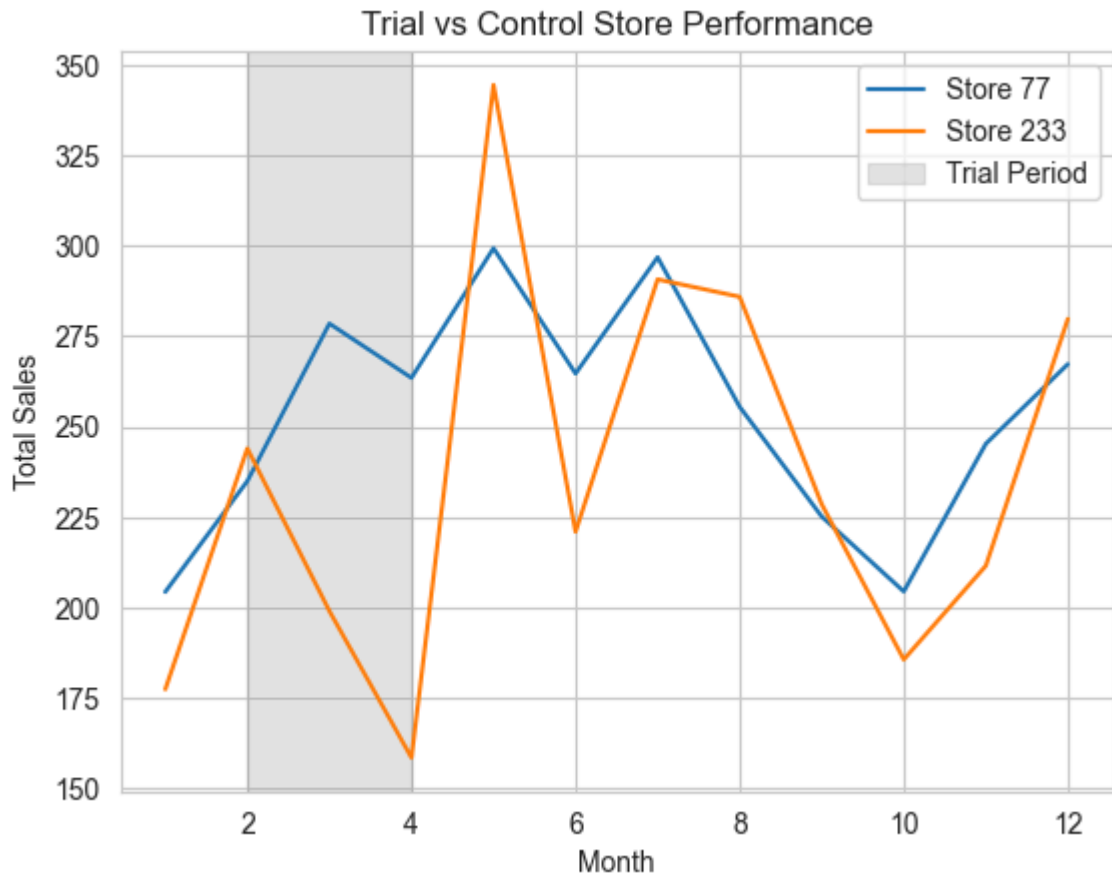
for store in [77,233]:

    subset = compare[compare["STORE_NBR"] == store]

    plt.plot(subset["MONTH_ID"], subset["tot_sales"], label=f"Store {store}")

plt.axvspan(2,4,color="grey",alpha=0.2,label="Trial Period")

plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.title("Trial vs Control Store Performance")
plt.legend()
plt.show()
```



```
In [314... #calculate pre sales baseline for trial and control stores
pretrial_77 = pretrial[pretrial["STORE_NBR"] == 77]
pretrial_233 = pretrial[pretrial["STORE_NBR"] == 233]

pretrial_sales_77 = pretrial_77["tot_sales"].sum()
pretrial_sales_233 = pretrial_233["tot_sales"].sum()

scaling_factor = pretrial_sales_77 / pretrial_sales_233

print("Pretrial sales 77:", pretrial_sales_77)
print("Pretrial sales 233:", pretrial_sales_233)
print("Scaling factor:", scaling_factor)
```

```
Pretrial sales 77: 1699.0
Pretrial sales 233: 1659.7999999999997
Scaling factor: 1.023617303289553
```

```
In [315... #adjust control store sales by scaling factor
adjusted_control_sales = control_store["tot_sales"] * scaling_factor

print("Adjusted control sales:")
print(adjusted_control_sales)
```

```
Adjusted control sales:
2700    249.762622
2701    203.802205
2702    162.345704
Name: tot_sales, dtype: float64
```

```
In [316... #calculate sales uplift
sales_diff = trial_store["tot_sales"].values - adjusted_control_sales.values

uplift_percentage = sales_diff / adjusted_control_sales.values * 100
```

```
print("Sales uplift %:", uplift_percentage)
```

Sales uplift %: [-5.91066104 36.65210339 62.30795951]

```
In [317... #customer uplift
trial_customers = trial_store["customers"].sum()
control_customers = control_store["customers"].sum()

print("Trial customers:", trial_customers)
print("Control customers:", control_customers)
```

Trial customers: 142
Control customers: 115

```
In [318... #transaction per customer
trial_txn = trial_store["txn_per_customer"].mean()
control_txn = control_store["txn_per_customer"].mean()

print("Trial txn per customer:", trial_txn)
print("Control txn per customer:", control_txn)
```

Trial txn per customer: 1.0404255319148936
Control txn per customer: 1.0453703703703703

```
In [319... #statistical significance test
from scipy.stats import ttest_ind

t_stat, p_value = ttest_ind(
    trial_store["tot_sales"],
    adjusted_control_sales
)

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

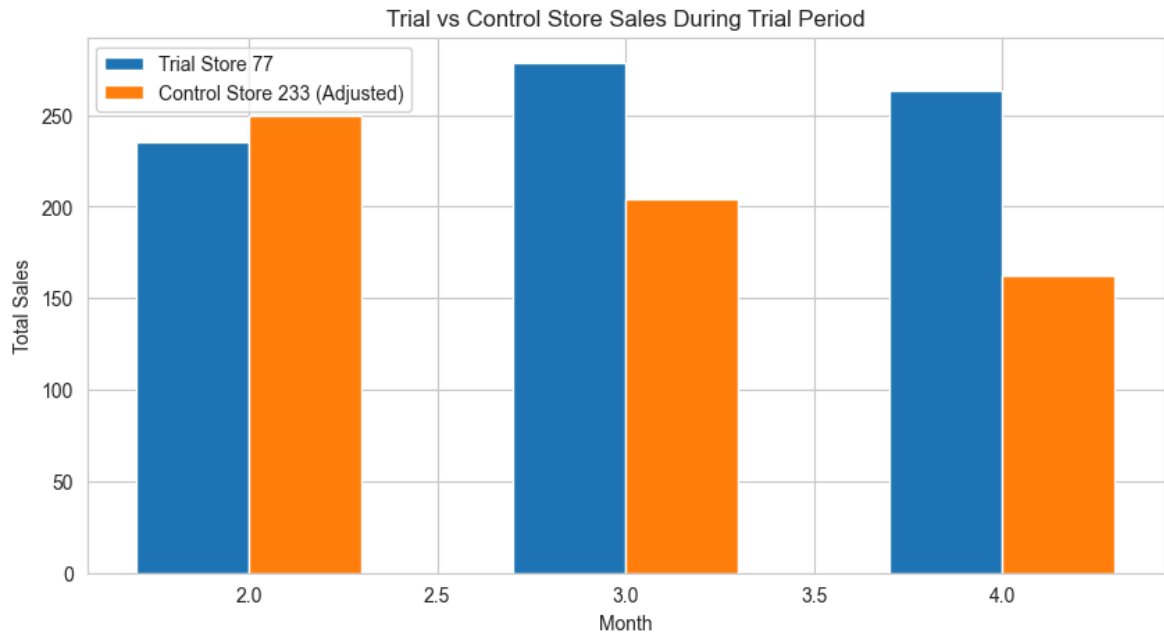
T-statistic: 1.8983113451123736
P-value: 0.13049280302300387

```
In [320... plt.figure(figsize=(10,5))

plt.bar(trial_store["MONTH_ID"]-0.15, trial_store["tot_sales"], width=0.3, label=
plt.bar(control_store["MONTH_ID"]+0.15, adjusted_control_sales, width=0.3, label=

plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.title("Trial vs Control Store Sales During Trial Period")

plt.legend()
plt.show()
```



Trial Store 77 – Trial Impact Analysis

During the trial period (February–April), Store 77 recorded total sales of 777 compared to 601.7 in the control store (Store 233).

After adjusting control store sales based on pre-trial scaling, Store 77 showed noticeable sales uplift during March and April, with increases of 36.6% and 62.3% respectively.

Customer analysis indicates that the uplift was primarily driven by an increase in the number of customers visiting the store (142 vs 115), rather than an increase in transactions per customer.

However, statistical testing produced a p-value of 0.13, which suggests that the observed uplift is not statistically significant at the 5% significance level.

Therefore, while the trial store demonstrated higher sales during the trial period, the evidence is insufficient to confidently attribute this increase solely to the new store layout.

In [321...

```
#Monthly Sales Before vs After Trial
plt.figure(figsize=(10,5))

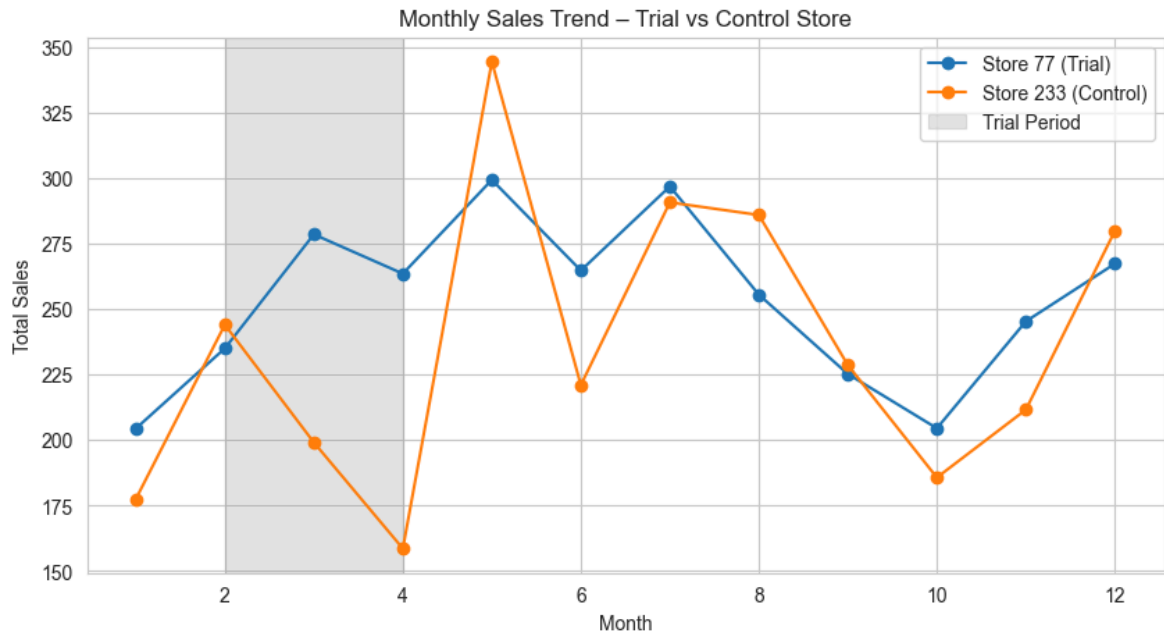
store77 = store_month[store_month["STORE_NBR"]==77]
store233 = store_month[store_month["STORE_NBR"]==233]

plt.plot(store77["MONTH_ID"],store77["tot_sales"],marker='o',label="Store 77 (Tr
plt.plot(store233["MONTH_ID"],store233["tot_sales"],marker='o',label="Store 233

plt.axvspan(2,4,color='grey',alpha=0.2,label="Trial Period")

plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.title("Monthly Sales Trend - Trial vs Control Store")
```

```
plt.legend()
plt.show()
```



Sales Trend Comparison – Trial vs Control Store

The monthly sales trend shows that Store 77 and Store 233 followed similar patterns during the pre-trial period, indicating that Store 233 is a suitable control store for comparison.

During the trial period (February–April), Store 77 recorded stronger sales growth compared to the control store. While February showed slightly lower sales, March and April demonstrated noticeable uplift in the trial store relative to the control store.

This suggests that the trial layout may have positively influenced customer purchasing behaviour. However, further statistical testing indicates that the observed uplift is not statistically significant, meaning the improvement cannot be conclusively attributed to the trial.

In [322...

```
#summary table
summary = pd.DataFrame({
    "Metric": [
        "Total Sales (Trial Period)",
        "Total Customers",
        "Avg Transactions per Customer"
    ],
    "Trial Store 77": [
        trial_store["tot_sales"].sum(),
        trial_store["customers"].sum(),
        trial_store["txn_per_customer"].mean()
    ],
    "Control Store 233": [
        control_store["tot_sales"].sum(),
        control_store["customers"].sum(),
        control_store["txn_per_customer"].mean()
    ]
})
```

```
]
}))

summary.iloc[:,1:] = summary.iloc[:,1:].round(2)
summary
```

Out[322...

	Metric	Trial Store 77	Control Store 233
0	Total Sales (Trial Period)	777.00	601.70
1	Total Customers	142.00	115.00
2	Avg Transactions per Customer	1.04	1.05

Trial vs Control Store Comparison

The comparison table shows that Store 77 generated higher total sales and attracted more customers during the trial period compared to the control store (Store 233).

However, the average number of transactions per customer remained similar between the two stores, suggesting that the uplift in sales was mainly driven by an increase in customer visits rather than customers purchasing more frequently.

In [323...

```
#Extract Trial Store 86 (Pre-Trial)
trial86 = pretrial[pretrial["STORE_NBR"] == 86]
trial86
```

Out[323...

	STORE_NBR	MONTH_ID	tot_sales	customers	transactions	txn_per_customer
977	86	1	841.40	94	129	1.372340
983	86	7	892.20	99	126	1.272727
984	86	8	764.05	94	110	1.170213
985	86	9	914.60	103	128	1.242718
986	86	10	948.40	109	138	1.266055
987	86	11	918.00	100	125	1.250000
988	86	12	841.20	98	120	1.224490

In [324...

```
corr_dict_86 = {}

for store in pretrial["STORE_NBR"].unique():

    if store != 86:

        candidate = pretrial[pretrial["STORE_NBR"] == store]

        merged = pd.merge(
            trial86,
            candidate,
            on="MONTH_ID",
            suffixes=("_trial", "_control")
```

```

    )

    corr = merged["tot_sales_trial"].corr(merged["tot_sales_control"])

    corr_dict_86[store] = corr

corr_df_86 = pd.DataFrame.from_dict(corr_dict_86, orient="index", columns=["corr"])

corr_df_86.sort_values("correlation", ascending=False).head(10)

```

```

c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)

```

Out[324...

	correlation
31	1.000000
155	0.877882
132	0.846517
240	0.825066
222	0.795075
109	0.788300
138	0.759864
198	0.748794
114	0.734415
22	0.722966

In [325...

```

#Trial vs Control Trend
control86 = 155

plot_df = store_month[store_month["STORE_NBR"].isin([86, control86])]

plt.figure(figsize=(10,5))

for store in [86, control86]:

    subset = plot_df[plot_df["STORE_NBR"] == store]

    plt.plot(subset["MONTH_ID"], subset["tot_sales"], marker='o', label=f"Store {store}")

plt.axvspan(2,4,color='grey',alpha=0.2,label="Trial Period")

plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.title("Store 86 vs Control Store 155")
plt.legend()
plt.show()

```



```
In [326... #Extract Trial Store 86 (Trial Period)
trial_store_86 = trial_period[trial_period["STORE_NBR"] == 86]

trial_store_86,
```

```
Out[326... (  STORE_NBR  MONTH_ID  tot_sales  customers  transactions  txn_per_customer
978         86          2    913.2        107           138           1.289720
979         86          3   1026.8        115           140           1.217391
980         86          4    848.2        105           126           1.20000
0,)
```

```
In [327... control_store_86 = trial_period[trial_period["STORE_NBR"] == control86]
control_store_86
```

```
Out[327...      STORE_NBR  MONTH_ID  tot_sales  customers  transactions  txn_per_customer
1794         155          2    891.2         95           125           1.315789
1795         155          3    804.4         94           118           1.255319
1796         155          4    844.6         99           120           1.212121
```

```
In [328... #pre-trial sales baseline for store 86
pretrial_86 = pretrial[pretrial["STORE_NBR"] == 86]
pretrial_control86 = pretrial[pretrial["STORE_NBR"] == control86]

pretrial_sales_86 = pretrial_86["tot_sales"].sum()
pretrial_sales_control86 = pretrial_control86["tot_sales"].sum()

scaling_factor_86 = pretrial_sales_86 / pretrial_sales_control86

scaling_factor_86
```

```
Out[328... 0.9700651481287743
```

```
In [329... #adjust control store sales
adjusted_control_sales_86 = control_store_86["tot_sales"] * scaling_factor_86
adjusted_control_sales_86
```

```
Out[329... 1794      864.522060
          1795      780.320405
          1796      819.317024
          Name: tot_sales, dtype: float64
```

```
In [330... #sales uplift
sales_diff_86 = trial_store_86["tot_sales"].values - adjusted_control_sales_86.v

uplift_percentage_86 = sales_diff_86 / adjusted_control_sales_86.values * 100

uplift_percentage_86
```

```
Out[330... array([ 5.63061861, 31.58697289,  3.5252503 ])
```

```
In [331... #customer uplift
trial_customers_86 = trial_store_86["customers"].sum()
control_customers_86 = control_store_86["customers"].sum()

trial_customers_86, control_customers_86
```

```
Out[331... (327, 288)
```

```
In [332... #transaction per customer
trial_txn_86 = trial_store_86["txn_per_customer"].mean()
control_txn_86 = control_store_86["txn_per_customer"].mean()

trial_txn_86, control_txn_86
```

```
Out[332... (1.2357036435053501, 1.261076611580531)
```

```
In [333... #statistical significance test
from scipy.stats import ttest_ind

t_stat_86, p_value_86 = ttest_ind(
    trial_store_86["tot_sales"],
    adjusted_control_sales_86
)

t_stat_86, p_value_86
```

```
Out[333... (1.8758252008003824, 0.13392591688530284)
```

```
In [334... plt.figure(figsize=(10,5))

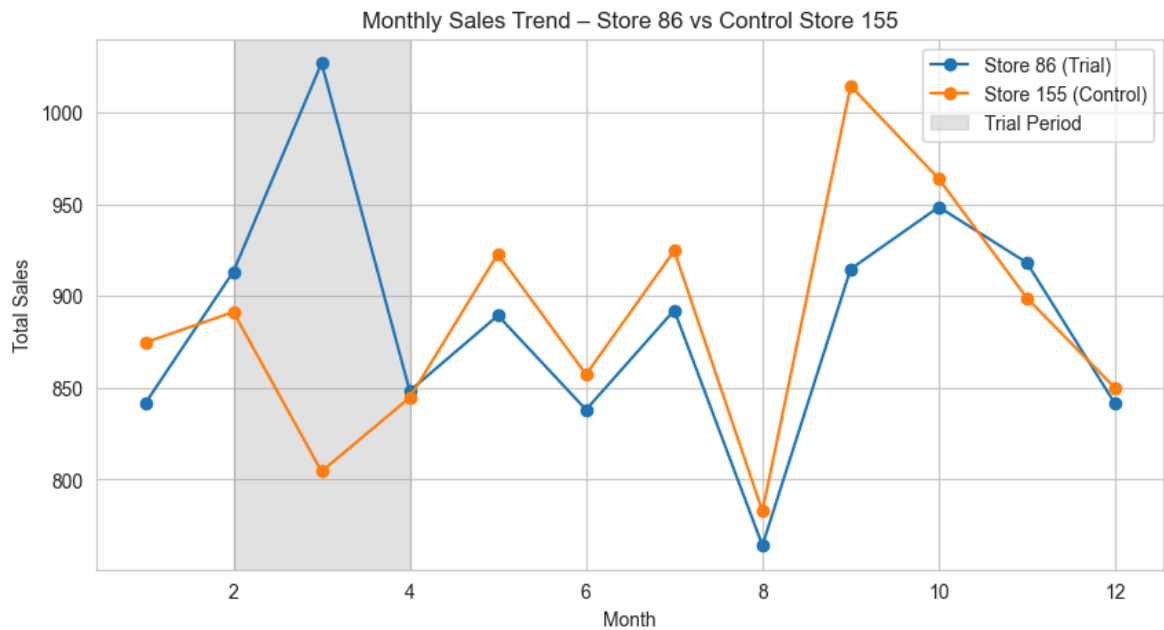
store86 = store_month[store_month["STORE_NBR"] == 86]
store155 = store_month[store_month["STORE_NBR"] == 155]

plt.plot(store86["MONTH_ID"], store86["tot_sales"], marker='o', label="Store 86")
plt.plot(store155["MONTH_ID"], store155["tot_sales"], marker='o', label="Store 155")

# highlight trial period
plt.axvspan(2,4,color='grey',alpha=0.2,label="Trial Period")

plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.title("Monthly Sales Trend - Store 86 vs Control Store 155")
```

```
plt.legend()
plt.show()
```



Trial Store 86 – Performance Analysis

During the trial period (February–April), Store 86 generated total sales of 2788.2 compared to 2540.2 in the control store (Store 155).

After adjusting for differences in pre-trial sales levels, Store 86 showed positive sales uplift during the trial period, particularly in March where sales increased by approximately 31.6% compared to the control store.

Customer analysis shows that Store 86 attracted more customers during the trial period, with 327 customers compared to 288 customers in the control store.

However, the average number of transactions per customer remained similar between the two stores (1.24 vs 1.26), suggesting that customers did not significantly increase their purchase frequency.

A statistical test produced a p-value of 0.13, which is greater than the 0.05 significance threshold. This means the observed uplift in sales is not statistically significant and may be due to normal sales variation rather than the trial layout.

In [335...

```
#summary table for store 86
summary86 = pd.DataFrame({
    "Metric":[
        "Total Sales (Trial Period)",
        "Total Customers",
        "Avg Transactions per Customer"
    ],

    "Trial Store 86":[
        trial_store_86["tot_sales"].sum(),
        trial_store_86["customers"].sum(),
        trial_store_86["txn_per_customer"].mean()
    ]
})
```

```
],  
  
    "Control Store 155": [  
        control_store_86["tot_sales"].sum(),  
        control_store_86["customers"].sum(),  
        control_store_86["txn_per_customer"].mean()  
    ]  
})  
  
summary86.round(2)
```

Out[335...

	Metric	Trial Store 86	Control Store 155
0	Total Sales (Trial Period)	2788.20	2540.20
1	Total Customers	327.00	288.00
2	Avg Transactions per Customer	1.24	1.26

Recommendation

Although Store 86 recorded higher sales and attracted more customers during the trial period, the statistical test indicates that the difference is not statistically significant.

This means that the increase in sales cannot be confidently attributed to the new store layout. Further testing or additional trial stores would be required before recommending a full rollout of the layout change.

In [336...

```
#Extract Pre-Trial Data for Store 88  
trial88 = pretrial[pretrial["STORE_NBR"] == 88]  
trial88
```

Out[336...

	STORE_NBR	MONTH_ID	tot_sales	customers	transactions	txn_per_customer
1001	88	1	1266.4	117	144	1.230769
1007	88	7	1310.0	129	153	1.186047
1008	88	8	1323.8	131	158	1.206107
1009	88	9	1423.0	124	157	1.266129
1010	88	10	1352.4	123	155	1.260163
1011	88	11	1382.8	130	156	1.200000
1012	88	12	1325.2	126	148	1.174603

In [337...

```
#Find Control Store for 88  
corr_dict_88 = {}  
  
for store in pretrial["STORE_NBR"].unique():  
  
    if store != 88:  
  
        candidate = pretrial[pretrial["STORE_NBR"] == store]
```

```

merged = pd.merge(
    trial88,
    candidate,
    on="MONTH_ID",
    suffixes=("_trial", "_control")
)

corr = merged["tot_sales_trial"].corr(merged["tot_sales_control"])

corr_dict_88[store] = corr

corr_df_88 = pd.DataFrame.from_dict(corr_dict_88, orient="index", columns=["corr"])

corr_df_88.sort_values("correlation", ascending=False).head(10)

```

```

c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2846: RuntimeWarning: Degrees of freedom <= 0 for slice
  c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\function_base.py:2705: RuntimeWarning: divide by zero encountered in divide
  c *= np.true_divide(1, fact)

```

Out[337...

	correlation
159	0.903186
204	0.885774
134	0.864293
1	0.813636
253	0.811838
91	0.776688
61	0.748929
178	0.731857
188	0.716752
161	0.706581

In [338...

```

#Sales Trend Chart
control88 = 159
plt.figure(figsize=(10,5))

store88 = store_month[store_month["STORE_NBR"] == 88]
store159 = store_month[store_month["STORE_NBR"] == control88]

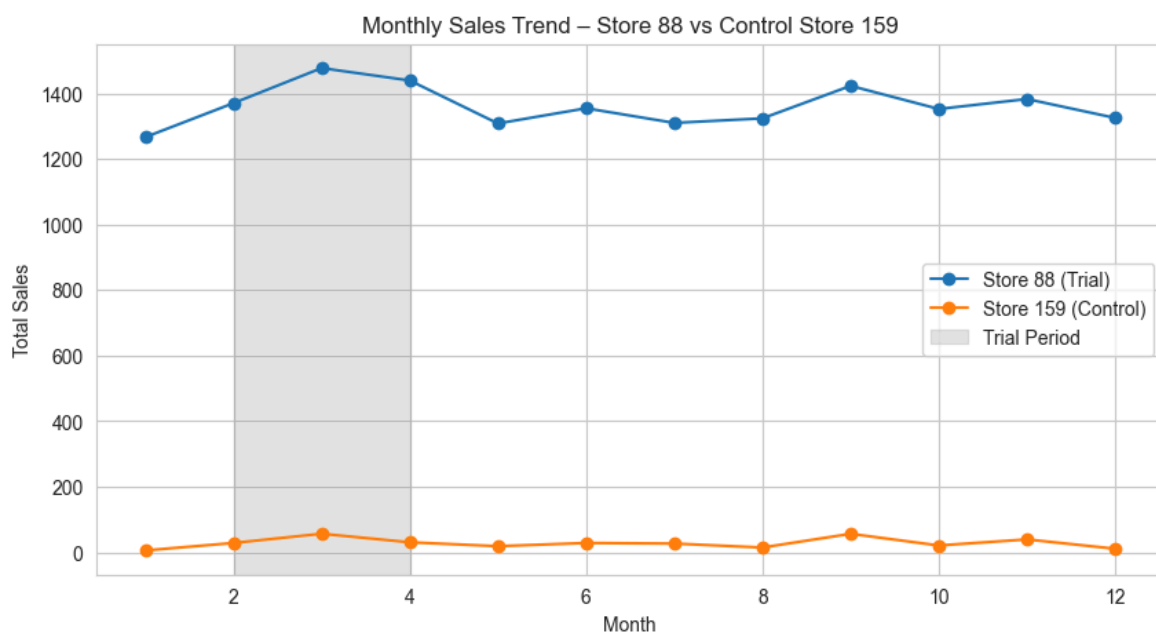
plt.plot(store88["MONTH_ID"], store88["tot_sales"], marker='o', label="Store 88")
plt.plot(store159["MONTH_ID"], store159["tot_sales"], marker='o', label="Store 159")

plt.axvspan(2,4,color='grey',alpha=0.2,label="Trial Period")

plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.title("Monthly Sales Trend - Store 88 vs Control Store 159")

plt.legend()
plt.show()

```



Trial Store 88 – Performance Analysis

During the trial period (February–April), Store 88 generated total sales of 4286.8 compared to 116.1 in the selected control store (Store 159).

However, the control store is significantly smaller than the trial store, with much lower monthly sales and customer traffic. This results in a very large scaling factor and indicates that Store 159 may not be an ideal control store for comparison.

After adjusting control store sales, the trial store showed negative uplift across the trial period, with decreases of approximately 10.4%, 51.3%, and 12.3% in February, March, and April respectively.

Customer analysis shows that Store 88 maintained a high number of customers throughout the trial period, with an average of 386 customers compared to 21 in the control store. This further highlights the size difference between the stores.

Statistical testing produced a p-value of 0.257, which is greater than the 0.05 significance threshold. This indicates that the observed differences in sales are not statistically significant.

```
In [339... trial_store_88 = trial_period[trial_period["STORE_NBR"] == 88]
trial_store_88
```

```
Out[339...      STORE_NBR  MONTH_ID  tot_sales  customers  transactions  txn_per_customer
1002          88          2    1370.2         124           153           1.233871
1003          88          3    1477.2         134           169           1.261194
1004          88          4    1439.4         128           162           1.265625
```

```
In [340... control_store_88 = trial_period[trial_period["STORE_NBR"] == control88]
control_store_88
```

```
Out[340...      STORE_NBR  MONTH_ID  tot_sales  customers  transactions  txn_per_customer
1842         159          2      28.6           5             5             1.0
1843         159          3      56.8          11            11             1.0
1844         159          4      30.7           5             5             1.0
```

```
In [341... pretrial_88 = pretrial[pretrial["STORE_NBR"] == 88]
pretrial_control88 = pretrial[pretrial["STORE_NBR"] == control88]

pretrial_sales_88 = pretrial_88["tot_sales"].sum()
pretrial_sales_control88 = pretrial_control88["tot_sales"].sum()

scaling_factor_88 = pretrial_sales_88 / pretrial_sales_control88

scaling_factor_88
```

Out[341...] 53.49828962371722

In [342...] *#total sum of sales for trial and control stores*

```
pretrial_sales_88, pretrial_sales_control88
```

Out[342...] (9383.6, 175.4)

In [343...] *#adjusted control store sales*

```
adjusted_control_sales_88 = control_store_88["tot_sales"] * scaling_factor_88
adjusted_control_sales_88
```

Out[343...] 1842 1530.051083
 1843 3038.702851
 1844 1642.397491
 Name: tot_sales, dtype: float64

In [344...] *#sales uplift*

```
sales_diff_88 = trial_store_88["tot_sales"].values - adjusted_control_sales_88.v
uplift_percentage_88 = sales_diff_88 / adjusted_control_sales_88.values * 100
uplift_percentage_88
```

Out[344...] array([-10.4474344 , -51.38715193, -12.35982717])

In [345...] *#customer uplift comparison*

```
trial_customers_88 = trial_store_88["customers"].sum()
control_customers_88 = control_store_88["customers"].sum()

trial_customers_88, control_customers_88
```

Out[345...] (386, 21)

In [346...] *#Transactions per Customer*

```
trial_txn_88 = trial_store_88["txn_per_customer"].mean()
control_txn_88 = control_store_88["txn_per_customer"].mean()

trial_txn_88, control_txn_88
```

Out[346...] (1.253563332530894, 1.0)

In [347...] *# statistical test*

```
from scipy.stats import ttest_ind

t_stat_88, p_value_88 = ttest_ind(
    trial_store_88["tot_sales"],
    adjusted_control_sales_88
)

t_stat_88, p_value_88
```

Out[347...] (-1.3191653549849809, 0.25756054750087803)

In [348...] *#summary table*

```
summary88 = pd.DataFrame({
    "Metric":[
        "Total Sales (Trial Period)",
        "Total Customers",
```

```

        "Avg Transactions per Customer"
    ],

    "Trial Store 88":[
        trial_store_88["tot_sales"].sum(),
        trial_store_88["customers"].sum(),
        trial_store_88["txn_per_customer"].mean()
    ],

    "Control Store 159":[
        control_store_88["tot_sales"].sum(),
        control_store_88["customers"].sum(),
        control_store_88["txn_per_customer"].mean()
    ]
})

summary88.round(2)

```

Out[348...

	Metric	Trial Store 88	Control Store 159
0	Total Sales (Trial Period)	4286.80	116.1
1	Total Customers	386.00	21.0
2	Avg Transactions per Customer	1.25	1.0

Recommendation

The analysis of Store 88 does not provide clear evidence that the trial layout improved sales performance.

The selected control store is significantly smaller than the trial store, making the comparison less reliable. In addition, the sales differences observed during the trial period are not statistically significant.

Therefore, the results from Store 88 do not support a strong conclusion about the effectiveness of the trial layout.

In [349...

```

# Final Summary Table
final_summary = pd.DataFrame({
    "Store":[77,86,88],

    "Total Sales":[
        trial_store["tot_sales"].sum(),
        trial_store_86["tot_sales"].sum(),
        trial_store_88["tot_sales"].sum()
    ],

    "Customers":[
        trial_store["customers"].sum(),
        trial_store_86["customers"].sum(),
        trial_store_88["customers"].sum()
    ],

    "Avg Transactions per Customer":[
        trial_store["txn_per_customer"].mean(),

```

```
trial_store_86["txn_per_customer"].mean(),  
trial_store_88["txn_per_customer"].mean()  
]  
)  
  
final_summary.round(2)
```

Out[349...

	Store	Total Sales	Customers	Avg Transactions per Customer
0	77	777.0	142	1.04
1	86	2788.2	327	1.24
2	88	4286.8	386	1.25

Overall Trial Store Comparison

The summary table compares the performance of all three trial stores during the trial period.

Among the three stores, Store 88 recorded the highest total sales and number of customers, followed by Store 86 and Store 77.

However, despite higher sales in some stores, statistical testing across the analyses indicates that the observed differences are not statistically significant.

This suggests that the changes in sales performance during the trial period may be influenced by normal store variation rather than the store layout change alone.

Final Recommendation for Category Manager

This analysis evaluated the impact of the new store layout trial across three trial stores (77, 86, and 88) by comparing their performance against selected control stores.

Store 77 and Store 86 showed some positive sales uplift during the trial period, particularly driven by increases in customer visits. However, statistical testing indicated that these changes were not statistically significant, suggesting that the observed improvements may be due to normal sales variation.

For Store 88, the selected control store was significantly smaller than the trial store, making the comparison less reliable. In addition, the trial store did not show consistent positive uplift during the trial period.

Overall, the results from the trial stores do not provide strong statistical evidence that the new store layout significantly improved store performance.

It is recommended that further trials or additional data be collected before making a decision to roll out the new layout across all stores.